

Julesh Shrestha

FULL STACK DEVELOPER

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📍 Kathmandu, Nepal · Remote (UTC+5:45)

Five years of TypeScript and JavaScript in production — NestJS and Node services, React and React Native clients, Postgres and MySQL underneath, on AWS. The last two years have been in compliance software, where an audit trail and correct access control matter as much as the feature itself. I also hold an MTech in AI and still write Python when a model has to sit behind an API. The work I do best is the unglamorous kind: permissions, slow queries, and the paths that only run when something fails.

SKILLS

Languages	TypeScript, JavaScript, Python, SQL
Backend	NestJS, Express, GraphQL, REST API design, Sequelize, Prisma, FastAPI
Frontend	React, React Native, TanStack Query, Vue 2 and 3, Vuetify
Data	PostgreSQL (including JSONB), MySQL, MongoDB, Redis, OpenSearch, ETL pipelines, index and query tuning
Infra	AWS (EC2, S3, RDS/Aurora, IAM, OpenSearch), Docker, Nginx, Cloudflare, GitHub Actions and GitLab CI
Security	JWT and bearer-token auth, API keys and signed webhooks, RBAC and row-level access, rate limiting, presigned uploads, dependency and CVE hygiene, audit logging
ML	TensorFlow/Keras, CNNs, vector search (FAISS), ONNX Runtime, LLM and RAG integration, pandas

EXPERIENCE

Full Stack Developer — Dimitra Technology

Oct 2024 – Present

- Built the due-diligence API in NestJS: it runs a deforestation check, generates the PDF a regulator reads, and records the result on Solana. Clients authenticate with per-user secret keys, and the chain write is async so a slow node can't hold up compliance reporting.
- Designed the farm access-control layer — grant, revoke, membership-scoped visibility — enforced in the query, so technicians and factory staff only ever load the farms assigned to them.
- Gave compliance teams a real audit trail: client-IP middleware and a metadata interceptor feeding paginated activity logs of who changed what, when.
- Modelled roles and permissions for buying-station and factory operations: first-class roles, activation-key onboarding, role resolution at login, a permission matrix in the admin panel, and duplicate-registration guards.
- Built the multi-tenant layer super-admins use across products — tenant store, routing, session isolation, switching — with carbon credits as the first adopter.
- Reworked file handling: presigned S3 uploads with image normalisation, correct content-type and disposition, sanitised filenames, and a Puppeteer service for agreements, ID cards and reports after the old PDF library kept dropping sockets.
- Fixed the slow paths — a missing index on a hot join, reference tables looked up on every request, an org-wide user query nobody needed. Several admin screens went from seconds to instant.
- Swapped sequential IDs in public traceability links for encoded external IDs decoded server-side, closing an enumeration path on a page anyone with the URL can open.

Freelance Full Stack Developer

Nov 2021 – Present

- Built the mobile API layer end to end — multi-device token auth, order timeline, notification inbox — plus the caching and indexing that keep list screens fast.

- Wrote the Python ingest pipelines: stage first, merge second, versioned history, safe to re-run after a failure. Milestone events arrive by signed webhook with a dedup key; a nightly scan catches whatever the webhook missed.
- Kept production healthy — PHP-FPM sized against real memory use, database contention traced, access logs turned into traffic reports, CDN caching and bot filtering with authenticated routes left uncached.
- Built most of the React Native app (Expo, TypeScript): file-based navigation, a React Query data layer with pagination and prefetch, one API client handling token refresh and 401 sign-out, English/Chinese localisation down to date formats and request headers, push notifications, and a biometric unlock flow written as a state machine so each native error code gets the right response instead of an endless prompt.
- Earlier clients: an async Python ETL for high-volume market data, portfolio and data-processing systems in NestJS, PostgreSQL and React, and the backend and schema for a bike-sharing prototype.

Software Engineer — Terakoya Academia

May 2021 – Oct 2021

- Wrote a TypeScript plugin for the Backstage developer portal that let people search GitHub and GitLab repositories by custom parameters.
- Added corporate proxy support to a self-hosted PeerTube instance so it could make outbound requests from inside a corporate network.
- Maintained the company's React, Django and MySQL sites and internal tools.

Web Development Intern — Bitsbeat IT Solutions

Aug 2020 – Nov 2020

- First professional exposure to web development — built small applications with JavaScript, Node and MongoDB, and learned how a backend is put together in a team.

EDUCATION

MTech, Artificial Intelligence — Kathmandu University

2021 – 2024

Thesis on segmenting Wilms tumours from CT scans, comparing UNet, DeepLabv3 and SegFormer over 10,000+ images. Best model reached 94.5% IoU.

BSc, Computer Science — Kathmandu University

2016 – 2020

Final-year project on age and gender classification from facial images, working with a limited dataset.